

Title

Blackjack Advantage AI: Risk of Ruin and Bankroll Optimization Tool

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1. Introduction

This project focuses on building a practical tool for analyzing blackjack advantage play using generative AI and simulation. In blackjack, advantage players rely on concepts such as expected value, variance, bankroll management, and risk of ruin. These calculations are typically complex and require either advanced spreadsheets or specialized software.

The goal of this project is to create a user-friendly application that allows players to input game conditions and receive a complete risk analysis. The tool uses Monte Carlo simulation to model thousands of blackjack sessions and estimate expected profit, bankroll growth, and the probability of losing the entire bankroll.

In addition to simulation, the project integrates generative AI using the OpenAI API to explain results in simple terms and provide recommendations. This makes the tool not only analytical but also educational.

2. Methodology

The project is built using Python and the Streamlit framework to create an interactive user interface. Users input key variables such as starting bankroll, minimum and maximum bet, hands per hour, number of hours played, and number of simulations.

A Monte Carlo simulation is used to model blackjack sessions. Instead of assuming a fixed advantage, the simulation generates a fluctuating true count for each hand using a normal distribution. The player's advantage is estimated using the formula:

$$\text{Estimated Edge} = -0.5\% + (\text{True Count} \times 0.5\%)$$

The simulation also includes realistic advantage play behavior:

- Players wong out when the true count is below a threshold
- Bet size increases as the true count increases
- The player only plays hands when the count is favorable

For each simulation, the bankroll is updated after every hand based on the bet size and outcome. The process is repeated across many simulations to generate a distribution of possible outcomes.

3. Results and Analysis

The simulation outputs several key metrics:

- Average ending bankroll
- Expected profit or loss
- Risk of ruin
- Distribution of final bankrolls
- Distribution of true counts

Results are visualized using histograms to show how outcomes vary across simulations. The risk of ruin is calculated as the percentage of simulations in which the bankroll reaches zero.

One limitation encountered was computational time. Running very large simulations (e.g., 500 hours with 1,000 simulations) significantly increases runtime. To address this, the number of simulations was adjusted to maintain a balance between accuracy and performance.

4. Generative AI Integration

The OpenAI API is used to generate explanations of the simulation results. After running the simulation, the system sends the input parameters and results to a language model.

The model returns a simplified explanation that includes:

- Interpretation of expected profit
- Explanation of risk of ruin
- Advice on bankroll management
- Discussion of variance and strategy

This integration enhances the usability of the tool by making complex statistical results easier to understand.

5. Challenges

Several challenges were encountered during development:

- Designing a realistic simulation of fluctuating true count
- Implementing wonging behavior and bet spread logic
- Managing computational performance for large simulations
- Integrating the OpenAI API correctly within the application

These challenges were addressed through iterative testing and refinement of the simulation model.

6. Conclusion

This project successfully demonstrates how generative AI and simulation can be combined to create a practical decision-making tool. The Blackjack Advantage AI application provides users with insights into bankroll risk, expected performance, and optimal play strategies.

The tool could be expanded in the future by incorporating more detailed blackjack rules, more accurate probability models, or real-time data integration. Overall, the project meets its objective of creating a functional MVP that applies generative AI concepts to a real-world problem.

7. GitHub Repository

<https://github.com/jahernandez3-code/blackjack-advantage-ai/tree/main>